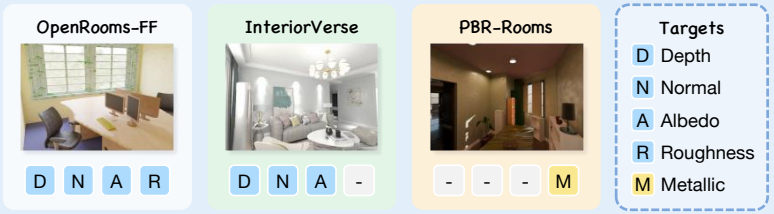
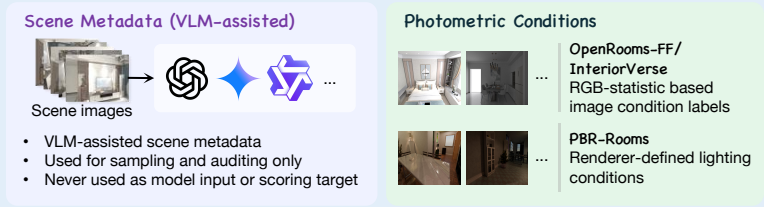
Source Eligibility & Split Construction

1 Renderer-Provided Dense Supervision



2 Sampling Metadata & Photometric Conditions

Different information sources are used for split construction, but none are provided to models.



Used only for benchmark split construction
(sampling, auditing, and defining photometric conditions)

3 Main- and Stress-Axis Construction

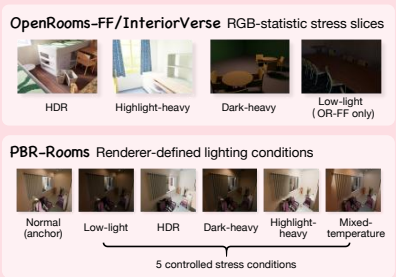
Main Axis (Retained Evaluation)

Scene-level retained subsets under target-specific eligibility.

Depth/ Normal/ Albedo	Source-balanced sampling from OpenRooms-FF and InteriorVerse.
Roughness	Sampled from OpenRooms-FF only.
Metallic	Sampled from PBR-Rooms only.

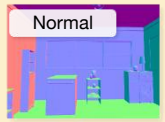
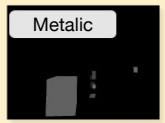
Stress Axis (Photometric Robustness)

Evaluate robustness under challenging illumination conditions.



Target-specific Probing Protocols

Each reported score corresponds to a fixed model-protocol pair.

Target	Access Setting	Key Instruction
	Single RGB image is used to generate a relative depth map under a fixed target-definition instruction and grayscale convention.	"Generate a relative depth map with a fixed grayscale convention. Preserve scene geometry and depth ordering."
	Single RGB image is paired with one fixed RGB-normal exemplar per run; three pre-specified runs define the official aggregate.	"Generate a surface normal map with the correct RGB convention. Preserve orientation and boundaries while suppressing texture and lighting."
	Single RGB image is used under a fixed albedo instruction. No illumination-analysis stage or auxiliary text is included in the official setting.	"Generate an intrinsic albedo map. Preserve structure and material colors while removing illumination effects."
	Single RGB image is used to produce a single-channel roughness map, excluding lighting and albedo effects.	"Generate a single-channel roughness map. Reflect surface roughness rather than shading or albedo."
	Single RGB image is used to produce a single-channel metallic map in [0,1], where brighter values indicate stronger metallicity.	"Generate a grayscale metallic map with values in [0,1]. Reflect metallic material presence rather than brightness or reflections."

Capability Diagnostics

Target	Diagnostic Comparisons
Depth	• Fixed prompt vs. RGB-depth exemplar • Fixed prompt vs. segmentation reference
Normal	• Full prompt + exemplars • Full prompt only • Minimal prompt + exemplars • Minimal prompt only
Albedo	• RGB-only official prompt vs. illumination-analysis variant • Full vs. weakened vs. minimal prompt • Minimal prompt vs. fixed exemplar
Roughness	• RGB-only prompt vs. segmentation soft prior • RGB-only prompt vs. segmentation region-fill • RGB-only prompt vs. fixed • RGB-roughness exemplar

Diagnostic Notes

- Used for cue-sensitivity analysis only; not part ofthe official comparison.
- Access diagnostics change the allowed cues, not the underlying target.
- Stress analysis evaluates robustness across challenging lighting conditions without changing model access.